$\Sigma F_y = 0$

$R_B - 80 - 60 + R_E \Rightarrow R_B + R_E = 140$

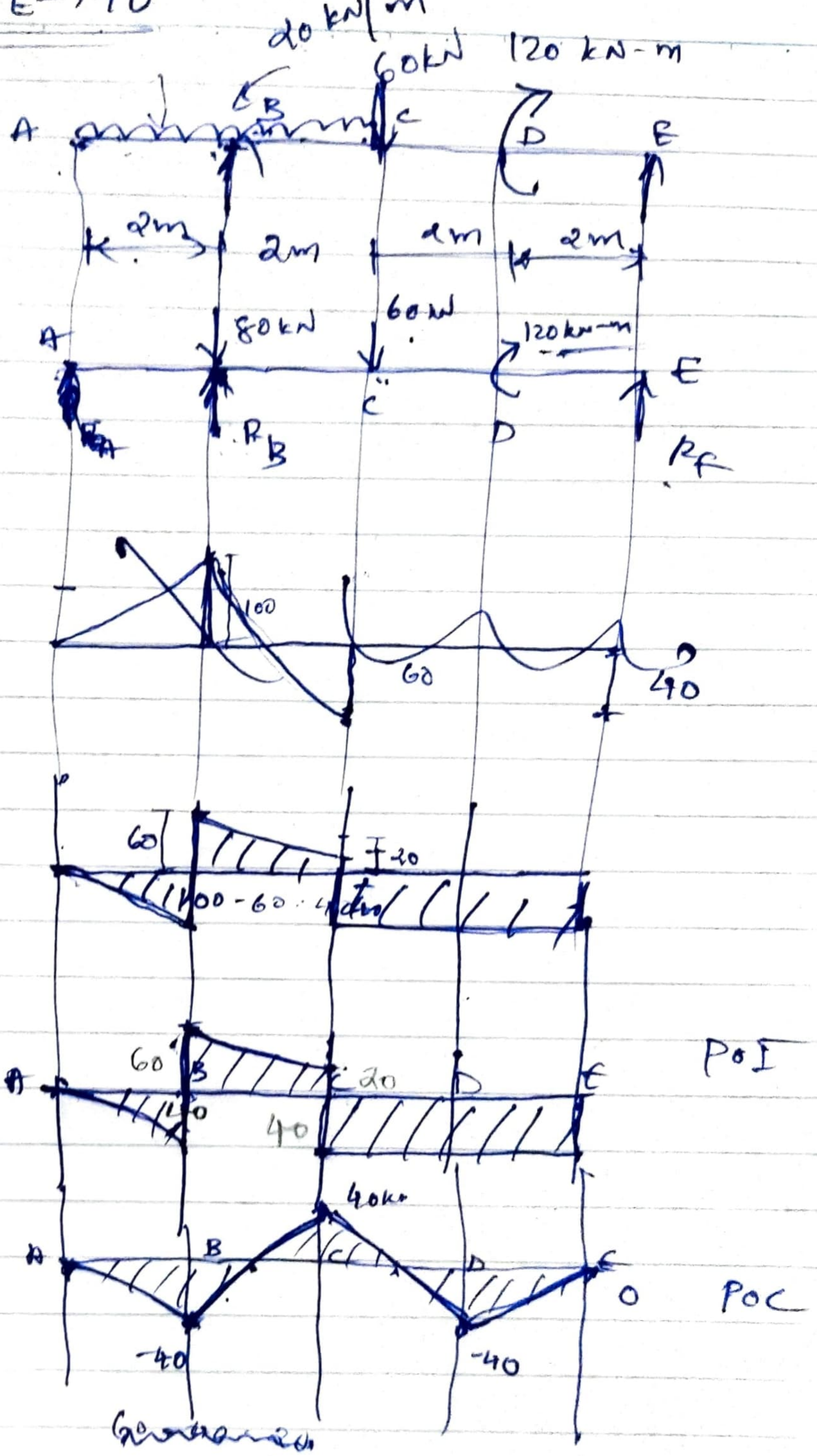
~~$R_B + R_E$~~

~~$R_A + R_B = -80 - 60$~~

~~$= 140\ kN$~~

$100 - 40 \quad = 40$

Date :



$\Sigma M_B = 0$

$60 \times 2 + 120 = R_E \times 6$

$\frac{240}{6} = R_E = 40\ kN$

$R_B + R_E = 140$

$R_B = 140 - 40$

$= 100\ kN$

S.F at A = NIL

S.F at B = $R_B = 100\ kN$

S.F at C = $-60\ kN$

S.F at D = NIL

S.F at E = $40\ kN$

$\Sigma M_A = 0$

$M_B = -20 \times 2 \times 1 = -40\ kN\text{-}m$

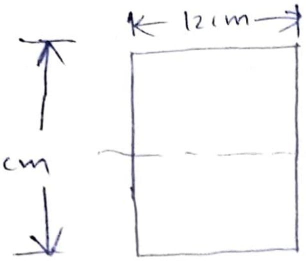
$M_C = -20 \times 4 \times 2 + 100 \times 2 = 40\ kNm$

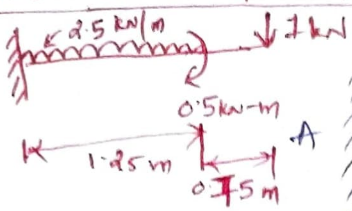
$M_D = -80 \times 4 + 100 \times 4 - 60 \times 2 = -40\ kNm$

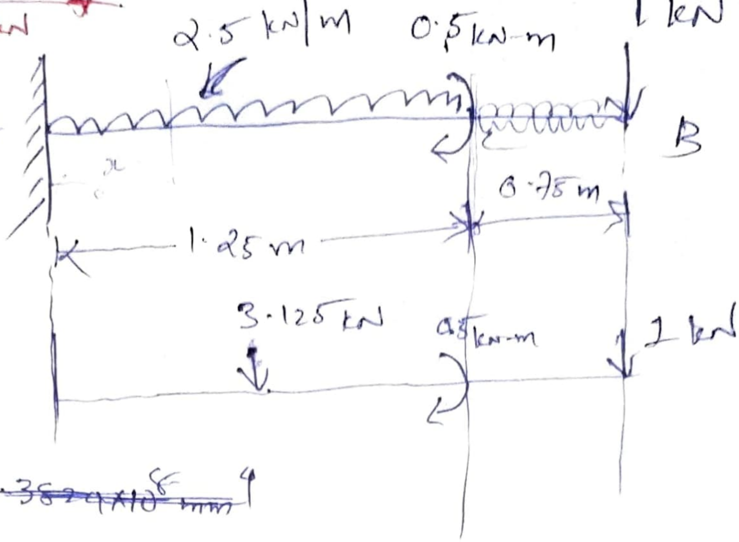
$M_E = -80 \times 6 + 100 \times 6 - 60 \times 4 + 120 = 0\ kN$

DEFLECTION OF CATILEVER WITH COUPLE

$l = 2m$







$E = 1 \times 10^4 \; N/mm^2$

$I = \frac{bd^3}{12} = \frac{12 \times 24^3}{12} = 13824 \, cm^4 = \cancel{1.3824 \times 10^8 \, mm^4}$

$R_A = 4.125 \, kN$

$$EI \frac{d^2y}{dx^2} = -2.5\frac{x^2}{2} \Big| + 2.5\frac{(x-1.25)^2}{2} + 0.5(x-1.25)^0$$

$x^0 = 1$

$$EI \frac{dy}{dx} = -2.5 \times \frac{x^3}{6} + c_1 \Big| + 2.5\frac{(x-1.25)^3}{6} + 0.5(x-1.25)$$

$$EI\, y = -2.5\frac{x^4}{24} + c_1 x + c_2 \Big| + 2.5\frac{(x-1.25)^4}{24} + 0.5\frac{(x-1.25)^2}{2}$$

B.C

I at $x = 0$; $y = 0$

$0 = c_2$

II ~~at x = 2; y =~~ @ max deflection ⇒ slope = 0

at $x = 2$; $\frac{dy}{dx} = 0$

$$0 = 2.5\frac{(2-1.25)^3}{6} + 0.5(2-1.25)$$

(PTO)

II $\boxed{\text{At max deflection ; Slope} = 0}$

ie at $x = 2$; $\frac{dy}{dx} = 0$

$$0 = -2.5 \times \frac{2^3}{6} + c_1 + 2.5\frac{(2-1.25)^3}{6} + 0.5\,(2-1.25)$$

$$= -\frac{10}{3} + c_1 + \frac{141}{256}$$

$$\therefore c_1 = 2.78$$

Deflection at B

$$EIy_B = -2.5 \times \frac{2^4}{24} + 2.78 \times 2 + 0 + 2.5\frac{(2-1.25)^4}{4} + 0.5\frac{(2-1.25)^2}{2}$$

$$EIy_B = 4.23\ m$$

$$y_B = \frac{4.23 \times 10^3}{EI} = \frac{4.23 \times 10^3}{1 \times 10^4 \times 13824 \times 10^4} = 3.059 \times 10^{-9}\ mm$$